

APPENDICES

A PREMISE GENERATION DETAILS

We curated a diverse set of seed premises spanning multiple thematic domains. To avoid redundancy and maintain structural coherence, we introduced a thematic classification framework informed by motifs from literature, philosophy, and film. The framework organizes premises into the following categories: **Existentialism/Philosophy**: addressing fundamental questions of self, existence, meaning, and cognition. **Socio-Political Issues**: focusing on themes of power, truth, morality, and social structures. **Psychological/Consciousness-Oriented Narratives**: exploring memory, dreams, identity, and perception. **Ethical Dilemmas**: presenting morally challenging scenarios that involve difficult trade-offs. **Time and Existence**: engaging with meta-physical settings such as temporality, fate, history, and immortality. Each seed premise is designed based on two core principles:

- (1) **Conflict or Reversal**. Every premise centers on a critical conflict or epistemic reversal, e.g., a philosopher discovers that their life’s work rests on fabricated evidence or a historian uncovers systematic manipulation of history. Such constructs encourage LLMs to explore diverse, high-stakes narrative trajectories.
- (2) **Concise Narrative Structure**. Premises follow a compact schema of subject + action/change + discovery/dilemma. Examples include a therapist experiences patients’ traumas as their own or a doctor discovers a cure that requires sacrificing one life to save many. This structure provides strong narrative hooks while remaining information-dense.

To avoid overly abstract or schematic seed premises, we expanded these seeds with additional narrative elements (e.g., characters, settings, temporal contexts). For example, a philosopher discovers that their life’s work is based on a fabricated premise was enriched into in a 19th-century European university town, a philosopher realizes that the foundation of their lifelong research rests on falsified historical records. This strategy preserves the thematic core while improving narrative specificity.

We further leveraged LLMs for lightweight rewriting, using structured prompts to generate stylistically varied variants. Post-processing then removed redundant or overly literal outputs, ensuring diversity and depth.

Through this procedure, we constructed an initial pool of 1,856 candidate premises. Example entries include “a scientist proposes a theorem proving that free will is an illusion and faces backlash from multiple sides” and “a therapist, after a medical accident, begins to experience their patients’ traumas as their own memories.” We then performed manual filtering to remove premises that were semantically empty, logically inconsistent, thematically redundant, or culturally sensitive. The resulting collection consists of 1,000 high-quality premises, each expressed in one to two sentences, semantically distinct, and sufficiently rich to drive story generation tasks.

B STORY GENERATION DETAILS

As shown in Table 3, we generate stories using LLMs as our candidate stories for bench construction. For the 533 LLM-generated premises, we adopt four LLMs to generate four candidate stories for each premise. For the 600 premises sampled from the WritingPrompts test set, we generate three candidate stories from three different LLMs, and include the original human-written story, also resulting in four candidates in total.

C HUMAN SCORING DETAILS

For the 600 instances containing one human-written story and three LLM-generated stories, we apply the preference ranking procedure in § 2.2 to rank only the LLM-generated stories. For the human-written stories,

we filter out very short stories, i.e., less than 100 words. Given the strict quality control of WritingPrompts with our additional filtering, we consider the human-written story as the highest-quality candidate among the four candidates. We conduct further verification and ask annotators to only verify whether the human-written story is indeed the best, without needing to judge the ranking of other stories. If the annotators are uncertain, they mark the instance as “unsure”. Finally, 91 instances are labeled as “unsure” and are dropped, resulting in 509 instances, each of which contains one human-written story and three LLM-generated stories. For the annotation of 230 instances with $\tau_{\text{avg}} < 0.6$, we conduct full human annotation, detailed in Section C.1. For the 394 instances with $\tau_{\text{avg}} \geq 0.6$, we conduct human verification. We first perform major voting, where we average the overall scores from the four LLMs to produce a final score for each story candidate. The average final scores result in the final ranking. Then, we provide the annotators with the premise and the ranked stories and ask them to confirm whether the higher-ranked stories are indeed better. If the annotator can not decide, we retain the original ranking. This human-verification process reduces annotation time by roughly 30% compared to full human annotation. The choice of the 0.6 threshold for τ_{avg} is based on an empirical heuristic: $\tau_{\text{avg}} = 0.6$ means approximately 80% concordant pairs and 20% discordant pairs, which we think is sufficiently high ranking agreement.

C.1 ANNOTATION INSTRUCTION

We implemented a local annotation platform to collect human judgments of story quality. We recruited and instructed annotators to evaluate candidate stories generated under the same premise. Annotators were instructed to read all candidate continuations for a given premise before assigning scores, and to ensure that their ratings clearly reflected differences in story quality. To maintain annotation reliability, we conducted periodic quality checks: after every 50 annotation groups, a random sample was reviewed to verify consistency and correctness.

C.2 ANNOTATION COORDINATION

We recruited students majoring in English to serve as annotators for this project. The demographic distribution of the annotators was 58% female and 42% male. All individuals involved in the annotation process held a bachelor’s degree. All staff members were compensated fairly based on an agreed-upon salary and workload. Formal contracts were signed with all annotators, and all employment practices adhered to local labor regulations. The privacy of the annotators was strictly protected, and no personal information was used for any purpose. The total cost for the annotation process, which included the annotation of stories, as well as the development and maintenance of the annotation platform, was approximately 10,000 USD.

D HUMAN-LLM BENCH DETAILS

Table 7 shows how we generate stories using LLMs to compare with human stories.

E TRAINING PAIRS GENERATION DETAILS

E.1 LLMs PAIRS GENERATION DETAILS

Table 3 shows prompt we use to generate story pairs with Llama3.1-8B-Instruct (Meta, 2024b) vs Llama-70B-Instruct (Meta, 2024a), DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI et al., 2025) vs DeepSeek-R1-Distill-Llama-70B (DeepSeek-AI et al., 2025), and Qwen2.5-14B (Bai et al., 2023) vs QwQ-32B (Qwen et al., 2025). On one hand we prompt larger LLMs to evaluate two stories generated by smaller LLMs, on the other hand we compare a story from a larger LLM with a story from a smaller LLM.

E.2 PREMISE BACK-GENERATION DETAILS

In addition to story rewriting and unguided generation, we also designed prompts (as shown in Table 4) that elicit premises from existing texts. Specifically, we provide two article titles and abstracts as input and ask the model to identify their shared elements, such as themes, settings, characters, or narrative developments, and condense them into a single premise. This premise is required to be sufficiently general yet logically consistent, such that both articles can be derived from it. This procedure allows us to systematically expand the premise pool with diverse and semantically grounded entries, while maintaining narrative coherence across generated stories.

E.3 REWRITING AS REJECTED STORY DETAILS

As shown in Table 6, to construct reliable preference pairs, we treat human-written stories as positive examples and generate low-quality negatives through controlled LLM rewriting. Specifically, we deliberately degrade text quality by altering character motivations, modifying key plot points, disrupting causal coherence, or simplifying language, thereby producing versions that remain broadly consistent with the original events but exhibit weaker narrative or stylistic qualities. These negatives are then paired with the human positives to form pairs for training the reward model. To ensure that the negatives are indeed of lower quality than the human versions, we performed random human checks on the generated outputs.

E.4 HUMAN-GUIDED CONTINUATION DETAILS

In fact, our initial method is to use the exact human-written stories as the chosen responses. However, the trained reward model performs poorly, with an average accuracy on StoryRMB below 30%. This may be due to that the distributional gap between human and LLM-generated stories is large (Tian et al., 2024), and the reward model can easily exploit superficial linguistic shortcuts rather than focusing on story quality. Therefore, we finally adopt the human-guided continuation approach, where the LLM continues from the human-provided context. We provide two representative prompt templates as shown in Table 5: (1) human-guided generation, where the model rewrites and extends a given story beginning while preserving its core elements; and (2) unguided generation, where the model creates an original story based solely on a title.

F DIMENSION SELECTION ALGORITHM

Algorithm 1 aims to categorize the candidate story set into a specific preferred dimension. It employs a hierarchical process to determine which dimension most highlights the chosen story. Firstly, during Stage 1, we compute the difference (mean_gap_d) between the chosen story’s score and the average score of all rejected stories, selecting the dimension with the largest gap as the preferred dimension. In Stage 2, if multiple dimensions yield gaps within the predefined tolerance τ (indicating a tie), we will compute the margin (margin_d) between the chosen story’s score and the highest score among the rejected stories, selecting the dimension with the largest margin to determine the preferred dimension. In Stage 3 should the tie persist, we will compute the variance (variance_d) of the rejected stories’ scores to assess the consistency of their performance. The dimension with the smallest variance is selected as the preferred dimension. Finally, if the analysis across the first three stages still results in a tie, the algorithm invokes the Ordinal Fallback (Stage 4), applying a pre-defined ranking to select the final preferred dimension.

G DATA STATISTICS

The data statistics of STORYRMB and the train dataset are shown in Table 9.